
EE 609: Network Information Theory

Course Code	EE 609
Course Name	Network Information Theory
L-T-P-C	3-0-0-3
Prerequisites	IC210, EE-304, Advanced Communication Theory
Intended for	4th year B.Tech. (CS+EE) and research students
Distribution	Elective for CS and EE
Approval	4th Senate (May 2013)

Course Contents

- **Module I:** Network information flow problem, Max-flow Min-cut Theorem, Point-to-Point Information Theory versus Network Information Theory. Brief overview of Point-to-Point Information Theory. [7 Lectures]
- **Multiple access channels:** Definition, Bounds on and single letter characterization of the capacity region, Time sharing, Gaussian MAC, extensions to more than two senders. [5 Lectures]
- **Broadcast channels (Degraded and General):** Definitions, Bounds on the capacity region, Superposition coding inner bound, Gaussian broadcast channel, Martons inner bound (also with common message), Outer bounds, Inner bounds for more than two receivers. [5 Lectures]
- **Channels with state:** Definition, Compound channel, Arbitrarily varying channel, Channels with random state, ISI channels, Dirty paper writing. [5 Lectures]
- **Distributed source coding:** Slepian-Wolf Theorem, Duality between Slepian-Wolf Theorem and MAC, Achievability schemes, Various generalizations (DSC with side-information, with helper, etc.). [5 Lectures]
- **Relay channels:** Definition, Cutset upper bounds, Gaussian relay channel, Performance of various achievability schemes (Decode-and-forward, Compress-and-forward, Compute- and- forward, Quantize-map-and-forward, Amplify-and-Forward, etc.). [5 Lectures]
- **Interference channels:** Definition, Coding schemes, Gaussian IC, Han-Kobayashi inner bound, Deterministic approximation of the Gaussian IC. [5 Lectures]
- **General multiterminal networks:** Gaussian networks, Capacity scaling-laws, Gupta-Kumar Network. [5 Lectures]

References

1. R. G. Gallager, **Information Theory and Reliable Communication**, Wiley, 1968.
2. A. El Gamal and Y.-H. Kim, **Network Information Theory**, Cambridge Univ. Press, 2012.
3. I. Csiszr and J. Krner, **Information Theory: Coding Theorems for Discrete Memoryless Systems**, Cambridge Univ. Press, 2011.
4. Related research papers.